



AI in Financial Risk Modelling

Can foundation models produce valid risk forecasts?



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Four ways to produce one number

- ▣ Tomorrow's 1% value-at-risk, from four kinds of model.
- ▣ A **statistical benchmark** you can compute by hand.
 - ▶ Sort the window, read off the quantile. No parameters at all.
- ▣ A **time-series foundation model**, pretrained elsewhere.
 - ▶ It has never seen this asset, and you cannot inspect what it has seen.
- ▣ A **commercial language model**, handed the returns as text.
 - ▶ No VaR head, no fitting: the number comes out of prompted generation.
- ▣ An **open-weights language model** you run yourself.
 - ▶ Three gigabytes, a laptop, and no key. This afternoon, you will.

The question is not which of them is cleverest

- ▣ It is whether you can tell, *from the output alone*, that the model answered the question you asked.

By the end of the session, you should be able to. . .

1. Distinguish a **point forecast**, a set of **marginal quantiles** and a **predictive distribution**, and say which one VaR needs.
2. Check whether a model or an API can produce the **risk functional you asked for** — before reading its numbers.
3. Separate **ranking**, **calibration testing** and **statistical power**, which answer three different questions.
4. Design an **as-of-valid** experiment with textual information and a pretrained model, with a control that identifies the effect you claim.

The empirical base

- ▣ S&P 500, 500 out-of-sample days, 2023-01-04 to 2024-12-30, $\alpha = 1\%$.
- ▣ Ten forecasts in the primary comparison plus three size diagnostics; five assets in the laboratory; one frozen protocol.

Act I

What is a risk forecast?

A risk model targets a different object

- A **forecasting** model targets the centre of tomorrow's distribution; a **risk** model targets a quantile of it.

$$\hat{r}_t = \mathbb{E}[r_t \mid \mathcal{I}_{t-1}], \quad \mathbb{P}(r_t < q_t(\alpha) \mid \mathcal{I}_{t-1}) = \alpha$$

- With $L_t = -r_t$ the loss and $q_t(\alpha) < 0$ the return quantile:

$$\text{VaR}_t(\alpha) = -q_t(\alpha) > 0, \quad \text{ES}_t(\alpha) = \mathbb{E}[L_t \mid L_t > \text{VaR}_t(\alpha)]$$

- VaR is a **threshold**; ES is an **average of the tail**.
 - ▶ VaR says nothing about what lies past it and is not sub-additive in general; ES is coherent (Acerbi & Tasche, 2002; Artzner et al., 1999), which is why Basel moved market-risk capital to 97.5% ES (Basel Committee on Banking Supervision, 2019).
- An excellent R^2 on the centre says nothing about the 1% quantile.
- We score VaR today: every model family here can be asked for it directly.
- Convention throughout: VaR and ES are **positive magnitudes**; figures are drawn in return space, where the same object is the negative threshold $q_t(\alpha)$.

Two factors, but only one is richly identified

$$q_t(\alpha) = \sigma_t \times q_\nu(\alpha), \quad \text{VaR}_t(\alpha) = -q_t(\alpha) > 0$$

The scale

- ▣ Observed and updated every day.
- ▣ Volatility clusters, so it is forecastable.
- ▣ A learning algorithm has data to learn from.

The shape

- ▣ At $\alpha = 1\%$: **five** events in 500 days.
- ▣ A 250-day HS window holds **two and a half** below the quantile.
- ▣ The tail index rests on those few observations.

- ▣ 1% is not 5% with a smaller number: it is the same question asked where the data runs out, and each method degrades differently on the way.
- ▣ Every result today says which of the two a given model got wrong.

Three questions people ask of one table

- **RANKING** — which forecast has the lower consistent loss?
 - ▶ A comparison. It never says either model is any good.
 - **VALIDITY** — is the forecast compatible with the target coverage and with the dynamics of the breaches?
 - ▶ An absolute test against a null, one model at a time.
 - **POWER** — would the test have detected a materially invalid model on this sample?
 - ▶ A property of the experiment, not of the model. Reported far less often than the other two.
 - **PIPELINE AUDIT** — did the object you scored come back from the question you asked?
 - ▶ Prior to all three, and the one Act III is about.
- Empirical frames from here carry the label of the question they answer.

Act II

The classical answers

The simplest answer: sort the window

- Historical simulation: sort the last W days, read off the empirical quantile.

$$\widehat{\text{VaR}}_t^{\text{HS}}(\alpha) = -\widehat{Q}_\alpha(r_{t-W}, \dots, r_{t-1})$$

- Nothing is fitted and nothing is assumed about the shape of the tail.
 - ▶ There is no parametric distribution to mis-specify. Window length and weighting remain modelling decisions.
- Every day in the window carries weight $1/W$.
 - ▶ Including one from four years ago, at full weight.
- It reacts only **discretely**, when observations enter or leave the window.
 - ▶ The response is coarse and window-dependent, and every day counts the same unless a weighted variant is used.

GARCH: model the scale, assume the shape

Engle (1982); Bollerslev (1986)

$$\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \widehat{\text{VaR}}_t(\alpha) = -(\mu + \sigma_t q_\nu(\alpha))$$

- The shock enters **squared**.
 - ▶ A -10% day and a $+10\%$ day move σ_t identically.
- The tail **family is imposed**; its parameters are estimated.
 - ▶ $q_\nu(\alpha)$ comes from the Student- t assumption, with ν fitted.
 - ▶ That is the scale/shape split made explicit in one equation.
- Half a century old, and still the benchmark machine learning must beat.

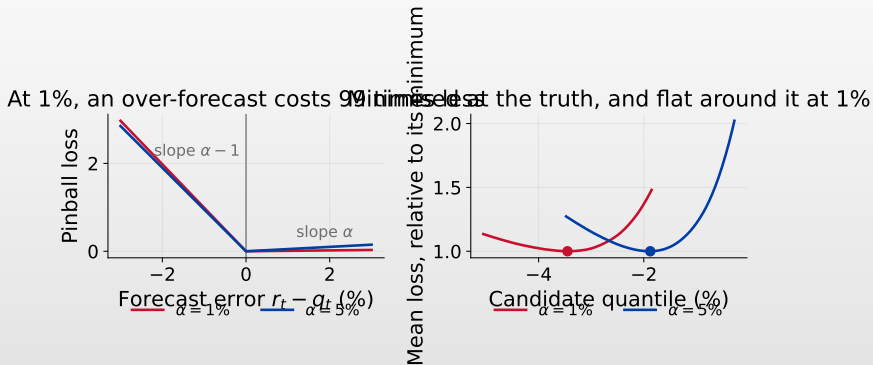
Quantile regression: learn the quantile end to end

- Minimise the **pinball** (check) loss (Koenker & Bassett, 1978) directly, with q_t the forecast quantile:

$$L_\alpha(r_t, q_t) = (\alpha - 1\{r_t < q_t\})(r_t - q_t)$$

- It is **strictly consistent** for the quantile.
 - ▶ Minimised in expectation at the truth, which makes it the right thing to rank on — and the breach count is not (Gneiting, 2011).
 - ▶ ES has no such loss of its own; the pair (VaR, ES) does, and it is in the appendix.
- Any function class drops in: linear, boosted trees, a neural net.
- The catch is where we are standing.
 - ▶ All observations enter the objective, but near 1% identification is weak and the surface is locally flat.
 - ▶ A flexible model fits the centre while the tail drifts.

The pinball loss, and why 1% is the hard case



Left: at $\alpha = 1\%$ the two arms have slopes 0.01 and -0.99 , so the loss punishes a threshold that is too close far harder than one too far. Right: the mean loss is minimised at the sample quantile, but at 1% every candidate within **0.64** points of it is inside 1% of the minimum — against 0.46 at 5%. Identification is weakest exactly where supervision looks.

Where can AI actually help?

- **The scale.** Thousands of training examples, a well-posed regression.
 - ▶ Genuine gains. A small network on log-volatility beats EWMA.
- **The shape.** A handful of events.
 - ▶ Flexibility here manufactures confidence, not information.
- **The joint law.** What a pretrained model claims to deliver.
 - ▶ Both factors at once, having never seen this asset. That is Act III.

-
- A flexible architecture cannot create additional realised tail events.
 - ▶ Any gain has to come from transferable structure, extra information or a modelling assumption — and each of those is checkable.

Act III

Time-series foundation models

Four words people use interchangeably

- **Supervised forecasting** — fit on *this* series, predict it.
- **Pretraining** — fit once, on a large corpus of *other* series.
- **Fine-tuning** — take a pretrained model, continue training on yours.
- **Zero-shot** — take a pretrained model, run it, change nothing.
 - ▶ Everything in this Act is zero-shot: no parameter has seen the S&P 500 in a way you control, and what it saw in pretraining you cannot inspect.
- Three outputs get reported as if interchangeable: a **point** forecast, **marginal quantiles** at fixed levels, a **predictive law**.
 - ▶ VaR needs the second *at your level*, or the third. A model that gives you quantiles on **its** grid has not given you yours.

Chronos: the series as a language

- ▣ Values are scaled and quantised into a fixed vocabulary of tokens.
- ▣ A sequence model is pretrained on a large corpus of time series.
- ▣ Forecasting is next-token prediction, decoded back to the value scale.

Two heads, and the difference is the whole Act

- ▣ **Chronos-T5** samples paths, so any quantile can be read off.
- ▣ **Chronos-Bolt** outputs a *fixed grid* of quantiles, far faster.

Ansari et al. (2024). TimesFM (Das et al., 2024) is the same idea with a different decoder.

 AIDA_aug_chronos

Why the input is prices

- In the tested pipeline, price **levels** are the compatible input representation.
 - ▶ Give it the last 512 closes, ask for the next.
- Recover the implied return:

$$r = 100 \log(\hat{P}_t / P_{t-1})$$

- A monotone transform.
 - ▶ So a price quantile maps to the return quantile of the same level.

Measured in the certified pipeline, S&P 500, 5541 out-of-sample days.

What happens if you feed returns

- Returns are near-zero-mean noise.
 - ▶ Predictive standard deviation **0.14** against a realised **1.22**, and the VaR collapses with it.

First question to any model: what can it answer?

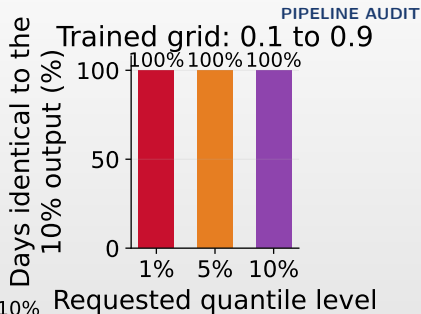
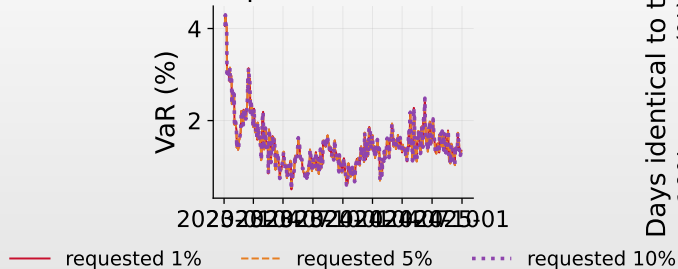
PIPELINE AUDIT

- Chronos-Bolt is a **quantile head**: it was trained on a fixed grid of levels and returns exactly those.
 - ▶ The trained grid is 0.1, 0.2, ..., 0.9. The lowest level it knows is 10%.
- Ask it for 1% and the call **succeeds**.
 - ▶ A warning is emitted, a number is returned, and the pipeline continues.
- So the capability check is one line, and it is not optional:
 - ▶ request several levels, then test whether the outputs actually differ.

- This is a property of the *artefact and its interface*, established before any forecast is scored.

You ask for the 1% quantile. You get the 10%.

Three requested levels, one line



Chronos-Bolt was trained on the grid 0.1, 0.2, ..., 0.9. Requests below 0.1 are clamped to the lowest trained level — on **100% of 500 days** the 1%, 5% and 10% outputs are bit-identical.

Why this is the important slide

PIPELINE AUDIT

- ▣ A pipeline that requested `quantile_levels=[0.01]` and took the number would have produced:
- ▣ a complete set of results, a leaderboard position, and a plot;
- ▣ no exception, no failed assertion, no missing value;
- ▣ a breach rate around 10% — **ten times** what it claims.

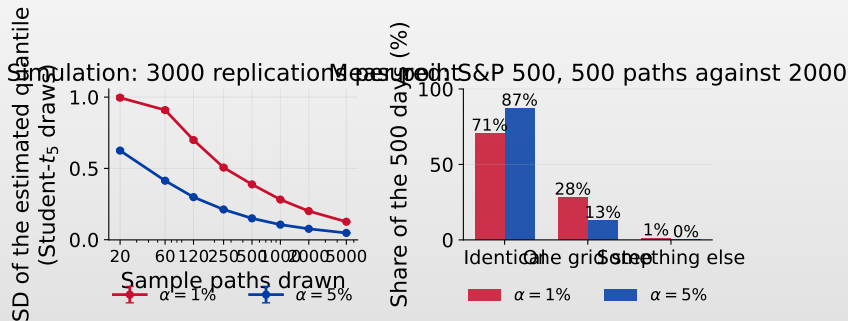
And the natural reading would have been wrong

- ▣ “The foundation model is badly calibrated in the tail” is a statement about the *pipeline*, dressed as a finding about the model.
- ▣ Ask what the artefact *can* answer, not only what you want.
- ▣ Then check the values, because the call succeeded.

The sampling head has a floor of its own

- Chronos-T5 reads the quantile off N sampled paths: at $N = 500$ and $\alpha = 1\%$, five draws lie below it.
- At the library default $N = 20$, a 1% quantile cannot be resolved below the $1/N$ Monte Carlo granularity.

PIPELINE AUDIT



Two runs on identical inputs, 500 against 2000 paths: the same VaR on 71% of days, **0.63** points apart on 28% — one step of the token grid.

Act IV

Language models as risk forecasters

Four different things people mean by “LLMs for risk”

1. **Feature extraction** — sentiment from text, fed to a conventional model.
 - ▶ The AI stops at a feature. The risk number is still econometrics.
2. **Event classification** — label the news flow.
3. **Probability elicitation** — ask for $\mathbb{P}(\text{tail event})$.
4. **Quantile elicitation** — ask for the VaR itself.

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- Only (3) and (4) put the language model *inside* the risk number.
 - ▶ That is what we test, following Pele et al. (2026).

Two ways to get a quantile out of a language model

Sampling — the published method

serialise $\rightarrow$ sample N completions $\rightarrow \hat{Q}_\alpha(\text{draws})$

- Statistically analogous to Chronos-T5 at the sampling-and-quantile stage: the quantile is an order statistic of the draws, and N sets the floor.

Elicitation — what we also run here

serialise $\rightarrow$ ask for $q_\alpha \rightarrow$ parse the reply

- One call instead of N , and it can be wrong in ways sampling cannot be: a positive quantile, an inverted pair.

Sampling follows Pele et al. (2026) and LLMTime (Gruver et al., 2023); the serialisation is theirs (base 10, prec 2, signed, ", " separator).

The prompt is a modelling decision

- Four information sets, and the differences between them are the experiment:
- `series` — the last 60 returns, **anonymised**.
 - ▶ No ticker, no dates, no price levels.
- `series+state` — the same, plus volatility, drawdown, the HS VaR.
- `dated` — the same returns, plus the asset name and the date.
- `dated+news` — the same, plus the real headlines available at $t - 1$.

Why two of them hide the date

- The model's pretraining covers these dates.
 - ▶ "S&P 500, 5 August 2024" invites recall, and recall is not available for tomorrow.

One rule decides whether the headlines mean anything

The as-of rule

- A headline may enter the prompt for date t only if it was published at or before the **as-of cutoff** on $t - 1$.
 - ▶ That cutoff is the asset's own market close.

- The cutoff is measured per exchange, not assumed.
 - ▶ S&P 20:00, DAX 15:30, Nikkei 06:00, gold 17:30, Bitcoin 24:00 UTC.
 - ▶ New York's close applied to the Nikkei would hand it fourteen hours of news published after Tokyo shut.

- Consecutive cutoffs **tile** the calendar.
 - ▶ No article falls in two windows and none falls in none. A fixed 24-hour window silently drops every evening and every weekend.

Why the control is dated

- Real headlines *reveal the period*.
 - ▶ They name companies, events and numbers that place the text in time.

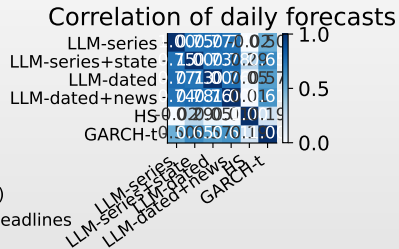
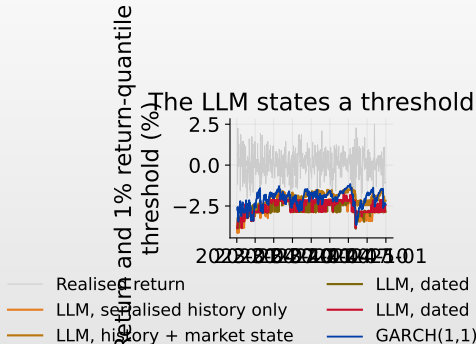
Configuration	period revealed	headlines	role
series	no	no	anonymised baseline
dated	yes	no	control
dated+news	yes	yes	treatment

- Comparing series with a news configuration would confound “the text carries information” with “the model now knows which period this is”.
 - ▶ Only the last two rows identify the first effect.

What we asked for, what came back, and what could have gone wrong

- One JSON object per day, **both levels in one reply**: a schema, so the reply can be checked rather than read.
- Five failure modes, in increasing order of danger:
 - ▶ It does not parse. It parses but the sign is wrong. The two quantiles are inverted. It parses, is correctly signed and ordered — and is constant. And it is right for the wrong reason, because it recognised the period.
 - ▶ Only the first announces itself.
- Across the four configurations, **2000 replies**: all parsed, none non-negative, none inverted.
 - ▶ Which settles the first three modes and says nothing about the last two.

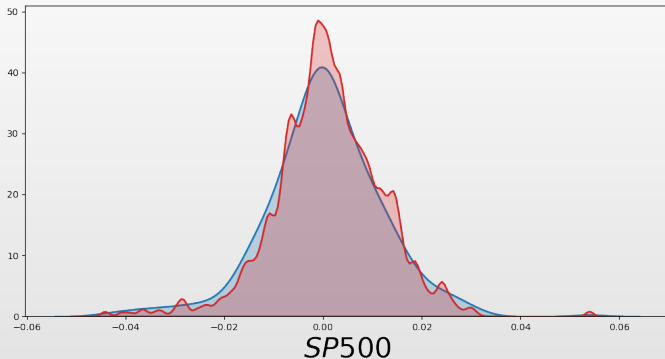
The LLM states a threshold, and it moves



RANKING

Correlation with GARCH- t is **0.50** for the anonymised configuration, and far lower with historical simulation. This is consistent with the language model primarily tracking **volatility** — the factor that has data.

What the sampled distribution looks like next to the truth



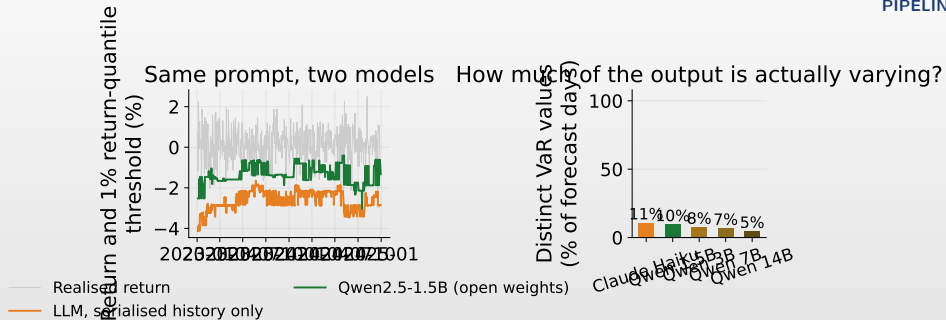
Blue: realised log returns. Red: the language model's draws, pooled over the test span. The draws are more concentrated at the centre than the returns are, and the difference sits in the shoulders rather than the far tail — the same finding as the threshold plot, read off the density instead of the path. Pele et al. (2026), GPT-4, context 120.

A model you can run on your own laptop

- A laboratory whose central exercise needs a credit card is a demonstration.
 - ▶ So the same prompt goes through **open weights**, downloaded from Hugging Face exactly the way Chronos is.
- Qwen2.5-1.5B-Instruct (Qwen Team, 2024) — about 3 GB and seconds per forecast, which is what a laptop can run.
 - ▶ This is the leg you run yourself this afternoon.
- The package also ships 3B, 7B and 14B, run over the same 500 days.
 - ▶ They answer one question: **does a larger model give a more informative VaR?** The next frame is the answer.
- The prompt, the schema and the parser are **imported** from the commercial stage.
 - ▶ So the legs differ in exactly one thing: which model reads the prompt.

In this setup, larger Qwen models give fewer distinct forecasts

PIPELINE AUDIT



The commercial model emits a distinct value on 10.8% of the 500 days; the open models, on 9.5% (1.5B), 7.6% (3B), 6.8% (7B) and 4.6% (14B). Every output is well formed, correctly signed and correctly ordered, and the **14B model moves its VaR on one day in twenty-two**. Conditional on this model family, this prompt, this decoding configuration and this elicitation protocol — it is not a scaling law.

The check nobody runs: count the distinct outputs

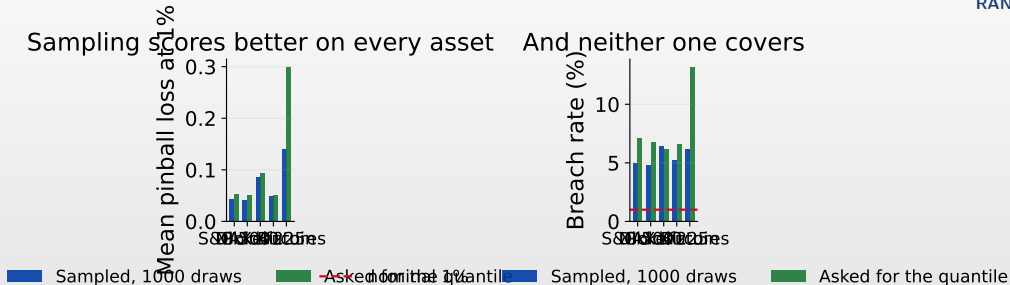
PIPELINE AUDIT

- ❑ Every validity test in the pipeline passes on every model in this section.
 - ▶ The reply parses, the sign is right and the two quantiles are in order.
- ❑ And the forecast moves on fewer than one day in ten.
 - ▶ Which no schema, no type check and no assertion can see.
- ❑ Only **counting distinct outputs** sees it.
 - ▶ One line of code, and it is in nobody's validation checklist.

- ❑ “Well formed” and “informative” are different properties.
 - ▶ The tooling only measures the first.

The same weights, asked two different ways

RANKING



One model, two extractions: Qwen2.5-1.5B asked for the number, against the same weights sampled **1000** times over the same 500 days, same parser, same scoring. Sampling wins on loss on **all five** assets and emits 15.8% distinct values against 9.5% on the S&P. Kupiec rejects **all ten** runs: breach rates of 4.8% to 13.2% against a nominal 1%.

Q AIDA_aug_local_llm

How far that comparison goes

- It is the only **fully controlled** experiment in the package.
 - ▶ The weights, the days and the parser are held fixed, so the only difference left is how the number was extracted.
- Sampling is the better extraction, and it is not close.
 - ▶ Lower loss on every asset, and roughly twice the distinct values.
- Neither is a valid 1% forecast at this model size.
 - ▶ Five breaches were expected. The best of these runs delivered 24.
- This is a statement about a 1.5B open model.
 - ▶ Pele et al. (2026) ran the sampling method on GPT-3.5 and GPT-4. Nothing here tests those.

Seven ways to produce one quantile

Family	Input	Native output	VaR from it	Hidden failure	First check
Historical simulation	Last W returns	Empirical law	Order statistic	Coarse, window-dated moves	Window length
GARCH- t	Returns	σ_t and assumed shape	$-\sigma_t q_\nu(\alpha)$	Tail family imposed	Shape assumption
Quantile regression	Features	The quantile itself	Direct	Flat objective near 1%	Identification
Chronos-Bolt	Price levels	Fixed quantile grid	Read the level	Clamped to the grid	Capability
Chronos-T5	Price levels	Sample paths	Order statistic	Resolution floor at $1/N$	Draw count
LLM, elicited	Serialised text	A stated number	Parse the reply	Well formed, near constant	Distinct values
LLM, sampled	Serialised text	N completions	Order statistic	The same floor, plus parsing	Draws and parser

Every row answers the same request differently, and every row fails quietly in its own way. The right-hand column is what you run *before* the backtest.

Act V

Validation

Backtesting in one line

- Count the days the threshold was crossed:

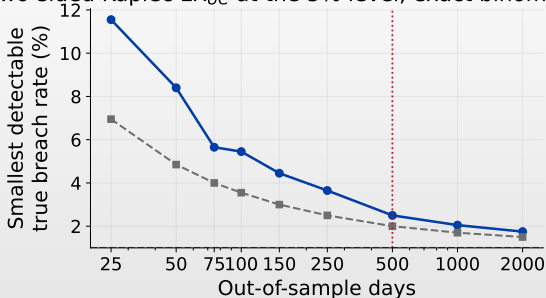
$$\hat{\alpha} = \frac{1}{T} \sum_{t=1}^T 1\{r_t < -\widehat{\text{VaR}}_t(\alpha)\}$$

- **Kupiec** tests whether $\hat{\alpha}$ differs from α (Kupiec, 1995).
 - ▶ $LR_{UC} \sim \chi_1^2$ under the null.
- **Christoffersen** (Christoffersen, 1998) tests whether the breaches *cluster*.
 - ▶ Right count, wrong timing is still a failure: breaches that arrive in one fortnight are the fortnight that matters.
- Their sum is conditional coverage, χ_2^2 .
 - ▶ Full likelihood-ratio expressions are in the appendix.

 AIDA_aug_backtest

How wrong a model must be before 500 days can see it

Two-sided Kupiec LR_{UC} at the 5% level, exact binomial power



—●— Detectable with 80% power

—■— Expected count alone enters the rejection region

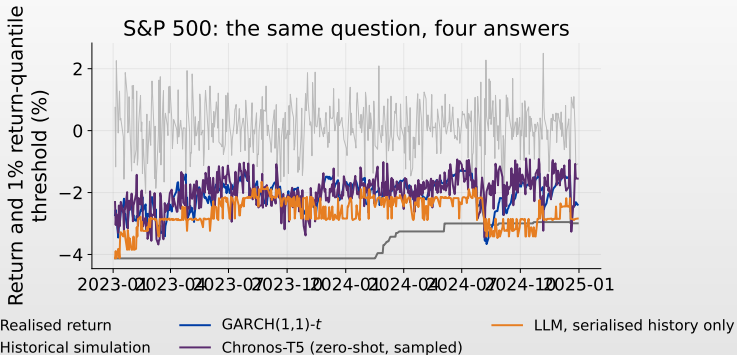
--- nominal 1%

... lab test span, 500 days

Two-sided Kupiec LR_{UC} at the 5% level, exact binomial power. Over 500 days, 80% power arrives only at a true breach rate of **2.50%**; over 5541 days, at 1.45%. The dashed line is the weaker criterion usually quoted, the rate whose *expected* count alone enters the rejection region.

 AIDA_aug_backtest

The same question, four answers



BANKING S&P 500, 500 out of sample days. Same forecast dates, same target, same scoring rule. The information representation differs by construction: history length and input form are not the same across HS, GARCH, Chronos and the LLM.

What the backtest says

VALIDITY

Model	n	Breaches	Rate (%)	UC	IND	Pinball
Neural volatility + t tail	500	3	0.60	pass	pass	0.0269
Claude Haiku 4.5, dated + headlines	500	3	0.60	pass	pass	0.0276
GARCH(1,1)- t	500	7	1.40	pass	pass	0.0277
Claude Haiku 4.5, series + state	500	6	1.20	pass	pass	0.0291
Claude Haiku 4.5, dated (control)	500	3	0.60	pass	pass	0.0298
Claude Haiku 4.5, series only	500	3	0.60	pass	pass	0.0299
Chronos-T5, 500 sample paths	500	6	1.20	pass	pass	0.0303
Historical simulation	500	2	0.40	pass	pass	0.0379
Qwen2.5-1.5B, sampled 1000×	500	25	5.00	reject	pass	0.0436
Qwen2.5-1.5B, asked	494	35	7.09	reject	pass	0.0533

Expected breaches: 5. Ten forecasts, one protocol.

Generated by `src/10_slide_tables.py`; source `precomputed/lab_backtest_SPX.csv`.

Eight of ten pass, and that says more about the test

POWER

- ❑ Breach counts of 2, 3, 6 and 7 all pass Kupiec at 1% on 500 days.
 - ▶ Five expected events cannot separate them. The test is not endorsing eight models; it is declining to speak.
- ❑ The two rejections miss by factors of five and seven.
 - ▶ 5.00% and 7.09% against a nominal 1%. It takes that much to be seen.
- ❑ So “passed the backtest” at this sample size means very little.
 - ▶ And it is exactly the sentence that appears in model documentation.
- ❑ Report the power alongside the verdict, or the verdict is decoration.

Does the S&P result replicate?

VALIDITY	Model	S&P 500	DAX 40	Nikkei 225	Gold futures	Bitcoin
	Neural volatility + t tail	3 (0.6%)	3 (0.6%)	4 (0.8%)	5 (1.0%)	2 (0.4%)
	GARCH(1,1)- t	7 (1.4%)	3 (0.6%)	6 (1.2%)	7 (1.4%)	1 (0.2%)
	Historical simulation	2 (0.4%)	0 (0.0%)	5 (1.0%)	7 (1.4%)	0 (0.0%)
	Chronos-T5, 500 sample paths	6 (1.2%)	11 (2.2%)	12 (2.4%)	16 (3.2%)	13 (2.6%)
	Claude Haiku 4.5, dated + headlines	3 (0.6%)	5 (1.0%)	3 (0.6%)	7 (1.4%)	4 (0.8%)
	Claude Haiku 4.5, series + state	6 (1.2%)	7 (1.4%)	7 (1.4%)	11 (2.2%)	7 (1.4%)
	Claude Haiku 4.5, dated (control)	3 (0.6%)	4 (0.8%)	3 (0.6%)	4 (0.8%)	5 (1.0%)
	Claude Haiku 4.5, series only	3 (0.6%)	3 (0.6%)	3 (0.6%)	2 (0.4%)	8 (1.6%)
	Qwen2.5-1.5B, asked	35 (7.1%)	34 (6.8%)	31 (6.2%)	33 (6.6%)	64 (13.2%)
	Qwen2.5-1.5B, sampled 1000×	25 (5.0%)	24 (4.8%)	32 (6.4%)	26 (5.2%)	31 (6.2%)
Days scored		500	500	500	500	500
Breaches expected		5	5	5	5	5

Breaches over 500 days, with the realised rate in brackets; red is a Kupiec rejection at 5%. **18 of the 50** model-asset cells are rejected. Both Qwen2.5-1.5B rows are rejected on all five markets, and Chronos-T5 passes on the S&P and is rejected on the other four. “Asked” means the model states the quantile in words; “sampled” means it is read off 1000 draws.

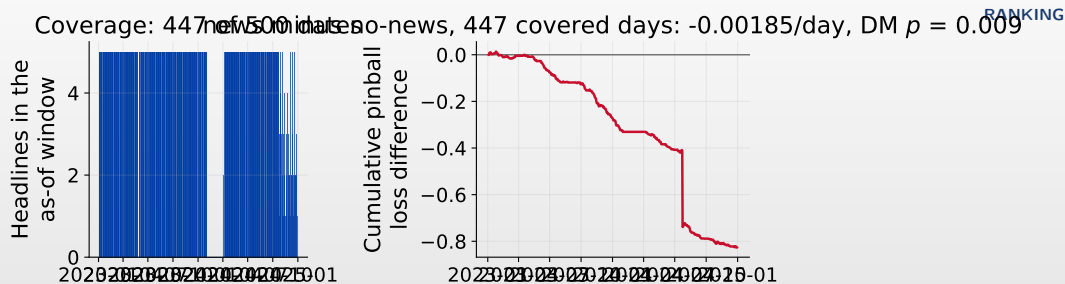
 AIDA_aug_backtest

Reading that table without fooling yourself

POWER

- Fifty tests at the 5% level.
 - ▶ Under a true null throughout, about **2 or 3** rejections are expected by chance alone. There are 18.
- The rejections are not scattered. They are concentrated.
 - ▶ Ten of them are the two open-weights rows, and four more are Chronos-T5.
- Chronos-T5 is the finding the single-asset table hid.
 - ▶ 1.2% on the S&P, then 2.2%, 2.4%, 3.2% and 2.6% elsewhere. One market's pass is not evidence.
- Historical simulation fails in the other direction.
 - ▶ Zero breaches on the DAX and on Bitcoin: too conservative to be rejected for being right.

Real news, at the level where it bites

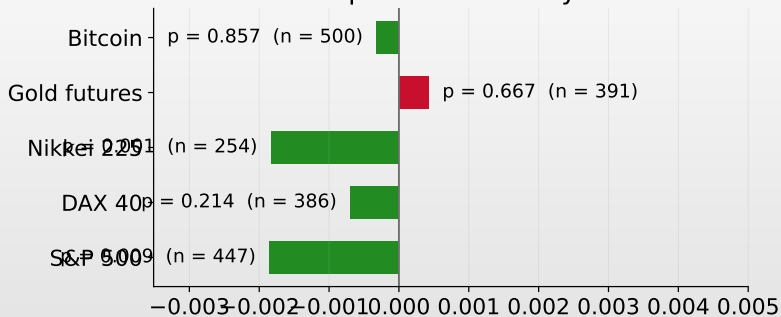


On the 447 scored days carrying at least one headline, the mean pinball loss falls by **0.00185** per day with news, Diebold–Mariano $p = 0.009$. At $\alpha = 5\%$ the same comparison gave $p = 0.61$.

Does it hold on the other four markets?

RANKING

The news null replicates on every market tested



Mean pinball loss, with headlines minus without (negative = headlines helped)

Significant on the S&P ($p = 0.009$) and the Nikkei ($p = 0.001$); not on the DAX, gold or Bitcoin.

 AIDA_aug_news

How far that result goes

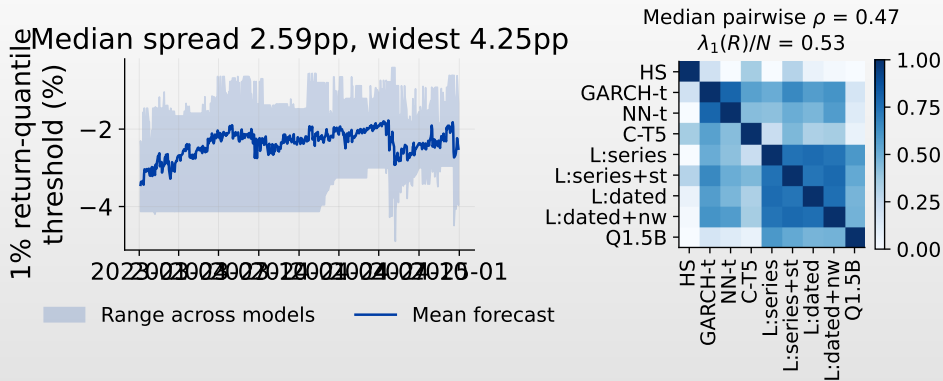
- It is **not** “news improves risk forecasts”.
 - ▶ It is: these headlines, at this horizon, at the 1% quantile, on two of five markets.
- **Five tests were run.** Bonferroni puts the threshold at 0.01.
 - ▶ The Nikkei survives it at 0.001; the S&P sits on the line at 0.009.
- **Contamination is not excluded.**
 - ▶ The prompt hides the date, but the regime is in the corpus regardless.
- **One model, one prompt, one calm period.**
 - ▶ 2023–2024 contains no 2008 and no March 2020.

Leakage: the failure that flatters every model

- **Window leakage** — the forecast day inside its own estimation window.
- **Alignment leakage** — a return series shifted one row against dates.
 - ▶ Silent, and it inflates every skill measure at once.
- **Selection leakage** — choosing the architecture on the reported period.
- **Temporal contamination** — the pretraining corpus contains the outcome.
 - ▶ The first three you can test for. The fourth you cannot, because you cannot read the corpus.

-
- Anonymising the prompt is therefore a design choice.
 - ▶ And the news alignment is audited rather than trusted.

How much of the disagreement is real?



Left: the band between the most and least conservative of the nine, day by day; median spread 2.59pp. A wide band is not evidence of independence, so the right panel measures it: median pairwise $\rho = 0.47$, and $\lambda_1(R)/N = 0.53$ against 0.11 under independence. The four LLM configurations form a block of their own at $\rho = 0.75$.

 AIDA_aug_backtest

The laboratory

Two hours in one notebook, and it runs without a key

What you will do this afternoon

- **Run selected forecasts yourself; validate the complete precomputed panel.**
 1. **Live, in full:** historical simulation you write, and the whole Chronos-Bolt clamp experiment. Both are seconds per day.
 2. **Live, on 5–20 dates:** the Chronos-T5 sampling head and the open-weights language model. Enough to see the mechanism and to check your output against the file.
 3. **From precomputed/:** the 500-day panels for every model and all five assets.
 4. **On the full panel:** backtests, DM tests, the news comparison against its control, and the cross-asset table.

Before you arrive

- Weights are cached in the room. Nobody spends the laboratory downloading 3 GB.
- Your group gets one of the five assets. The ranking is not stable across them, and the reason is the point.

Five things to leave with

1. A risk forecast is a **statistical functional**, not a plausible number.
 - ▶ Ask which functional, at which level, from which information set.
2. A foundation-model interface can silently return a **different object** from the one you requested.
 - ▶ The call succeeds, the plot renders, and the level is wrong. Check capability first.
3. At the 1% tail, **ranking**, **calibration** and **power** answer three different questions.
 - ▶ Eight of ten forecasts “passed” here, on five expected breaches.
4. Text adds information only under an **as-of-valid, controlled** design.
 - ▶ Headlines reveal the period, so the dated run is the control that makes the comparison mean anything.
5. Reproducibility is a property of the **whole pipeline**.
 - ▶ Data, prompt, model version, decoding, parser and validation — with open weights when the endpoint may change under you.

Can foundation models produce valid risk forecasts?

They produce *forecasts* readily, and they will answer any question you ask.

Validity is a separate finding, and it does not follow from plausibility or from a successful API call.

You need a **capability check**, an **output audit** and a **statistically appropriate validation** — in that order.

- Every number in this deck is regenerated by one script and re-derived from the data by another.

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Every DOI resolved and every link checked on 2026-08-02. Titles are live links.

Appendix

Derivations, diagnostics and the conventions this course assumes

One page of conventions this course assumes

Signs and objects

- ▣ r_t is the log return in %; the loss is $L_t = -r_t$.
- ▣ $q_t(\alpha) = F_t^{-1}(\alpha) < 0$ is the return quantile.
- ▣ $\text{VaR}_t(\alpha) = -q_t(\alpha) > 0$: a positive magnitude.
- ▣ $\text{ES}_t(\alpha) = \mathbb{E}[L_t \mid L_t > \text{VaR}_t(\alpha)]$.
- ▣ Breach indicator: $I_t = 1\{r_t < -\text{VaR}_t(\alpha)\}$.

Scale and shape

- ▣ $q_t(\alpha) = \sigma_t q_\nu(\alpha)$: a scale you observe daily, a shape identified by few events.

The pinball loss

- ▣ $L_\alpha(r, q) = (\alpha - 1\{r < q\})(r - q)$.
- ▣ Minimised in expectation at the true quantile, so it ranks forecasts.
- ▣ At $\alpha = 1\%$ the arms have slopes 0.01 and -0.99 .

How many breaches to expect at 1%

- ▣ 250 days: 2.5 expected.
- ▣ 500 days: 5 expected.
- ▣ Which is the whole reason Act V spends time on power.

What makes a loss the right loss?

Definition 1 (Consistency)

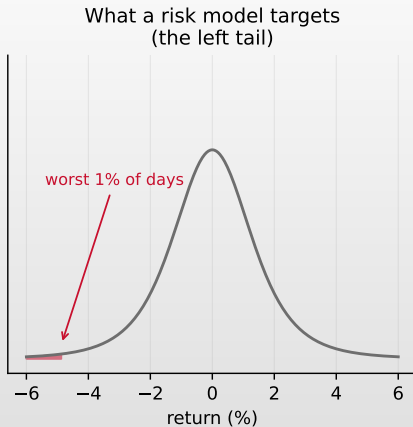
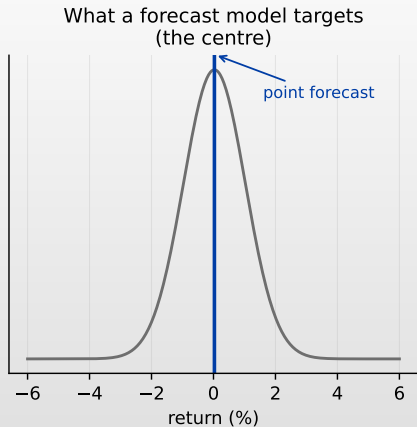
A scoring function L is **consistent** for a functional T if $\mathbb{E}[L(Y, T(F))] \leq \mathbb{E}[L(Y, x)]$ for every F and every report x . It is **strictly** consistent if equality forces $x = T(F)$.

Definition 2 (Elicitability)

T is **elicitable** if some strictly consistent L exists for it. Then $T(F) = \arg \min_x \mathbb{E}[L(Y, x)]$, and models can be ranked by average loss.

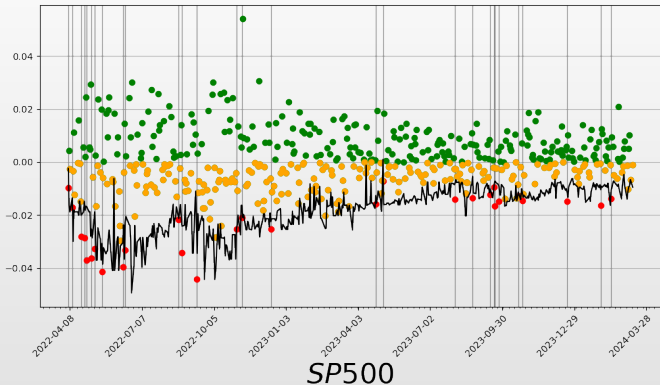
- ▣ Quantiles are elicitable; the pinball loss does it (Gneiting, [2011](#)).
- ▣ ES is *not* elicitable on its own — which is Act V's problem, not this one.

Point forecast, or the whole predictive distribution?



A risk model must deliver a distribution, or at least one quantile of it. Point accuracy is not the currency.

The published study, on the S&P 500



Pele et al. (2026), GPT-4, sampling with a 120-day context. Black is the VaR path, green and orange the returns above it, red the breaches. One of nine asset panels in the paper: same prompt, same parser, nine markets — the design this course reruns on open weights.

What the feed gave us

EODHD, S&P 500 index feed

- 35 293 articles fetched
- 2 128 survive the as-of rule
- 447 of 500 dates covered (89%)
- 4.8 headlines per covered date
- 0 published after their cutoff

- Median publication lead before the cutoff: **1.1 hours**.
 - ▶ The text is genuinely fresh, and genuinely prior.

The first symbol was wrong

- SPY.US returned 222 articles for two years.
- It left **67%** of dates empty.
- The feed was chosen on measured coverage, not plausibility.

Coverage and ranking, side by side



Expected 5 breaches. Historical simulation delivers 2, GARCH- t 7, and the open-weights model 35.

 AIDA_aug_backtest

Kupiec and Christoffersen in full

- With T days, x breaches and n_{ij} transitions between breach and no-breach:

Definition 3 (Unconditional coverage)

$$LR_{UC} = -2 \log \frac{(1 - \alpha)^{T-x} \alpha^x}{\left(1 - \frac{x}{T}\right)^{T-x} \left(\frac{x}{T}\right)^x} \sim \chi_1^2$$

(Kupiec, 1995)

Definition 4 (Independence)

$$LR_{IND} = -2 \log \frac{(1 - \hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi_1^2$$

(Christoffersen, 1998)

- $\hat{\pi}_{i1} = n_{i1}/(n_{i0} + n_{i1})$ is the breach probability after state i ; $\hat{\pi}$ pools them.
 - Conditional coverage is the sum of the two, χ_2^2 .

Comparing two models on the same days

Definition 5 (Diebold–Mariano)

With $d_t = L_\alpha(r_t, q_t^A) - L_\alpha(r_t, q_t^B)$,

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}} \xrightarrow{d} \mathcal{N}(0, 1), \quad H_0 : \mathbb{E}[d_t] = 0$$

(Diebold & Mariano, 1995)

- $\hat{\Omega}$ is a **HAC** estimate (Newey & West, 1987).
 - ▶ Loss differentials are serially dependent; the naive variance overstates significance.
- The test is paired, so both models must be scored on the *same* days.
 - ▶ A model that skips days is not comparable until the sample is aligned.
- Report $\bar{d}$ and its interval, not only the verdict.

Scoring ES means scoring the pair

Definition 6 (FZ_0 loss)

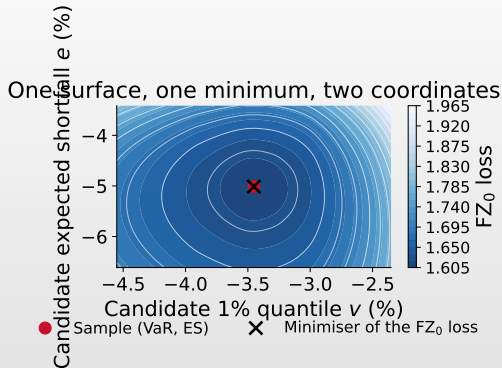
For a quantile $v < 0$ and a shortfall $e < 0$ at level α ,

$$L_{FZ_0}(r, v, e) = -\frac{1}{\alpha e} 1\{r \leq v\}(v - r) + \frac{v}{e} + \log(-e) - 1$$

(Fissler & Ziegel, 2016; Patton et al., 2019)

- ES is not elicitable on its own.
 - ▶ No scoring function is minimised by the true ES alone, so ES cannot be ranked the way a quantile can.
- The *pair* (VaR, ES) is jointly elicitable.
 - ▶ Which is why regulation can still be backtested (Nolde & Ziegel, 2017).

The minimum sits at the truth in both coordinates



S&P 500 test span. The minimiser of the FZ_0 loss lands on the sample pair in both coordinates, $(-3.45, -5.01)$. The valley is far shallower along ES than along VaR: the tail mean is estimated from the five days below the threshold, and the loss surface knows it.

SYNCRISK: when every desk runs the same model

$$D_t = \max_i \widehat{\text{VaR}}_{i,t} - \min_i \widehat{\text{VaR}}_{i,t}, \quad \rho_{ij} = \text{corr}(\widehat{\text{VaR}}_i, \widehat{\text{VaR}}_j)$$

- If every desk runs a similar model, every desk de-risks the same morning.
- Shared **pretraining** is a new channel for that.
 - ▶ The same weights behind many vendors, and no way to tell from outside.
- Four prompts are not four opinions: their median ρ is 0.75.
 - ▶ Historical simulation is the least correlated with the rest, at 0.21.
- Measuring ρ_{ij} across your own model set is a supervisory question.
 - ▶ You compute both quantities this afternoon.